

Edgar Rodríguez-Huerta

RIGHTS LAB SENIOR RESEARCH FELLOW IN SOCIAL SUSTAINABILITY AND COMPLEX SYSTEMS

Data Science, Sustainability and Supply Chain Mapping | £2.1m in Co-I & PI grant funding | FWCI: 1.42

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Experience

Rights Lab Senior Research Fellow in Social Sustainability and Complex Systems

University of Nottingham, Rights Lab

Nottingham, UK

Oct 2024 - to date

- Quantitative research focused on understanding climate change and decent work interconnections.
- Supply chain mapping and analysis to generate risk assessments using Social-LCA.
- Generation of climate change models to explore effect on labour conditions.

Rights Lab Research Fellow and Lead in Social-Ecological Systems Modelling

University of Nottingham, Rights Lab

Nottingham, UK

Dec 2021 to Sep 2024

- Awarded Global Talent Visa by UK Government with British Academy endorsement

Rights Lab Research Fellow in Sustainable Ecosystems and Society

University of Nottingham, Rights Lab

Nottingham, UK

Dec 2021 to Sep 2024

Industry Experience

Senior Project Developer and Business Intelligence Specialist

JLC Corporation

Mérida, México

Jan 2010 - Jan 2013

- Managed Business Intelligence and Enterprise Resource Program software, including stakeholder training
- Designed dashboards and reports for retail, financial and academic sectors.

Education

Ph.D. in Sustainability Science

Universitat Politècnica de Catalunya, Institute for Sustainability Science

Barcelona, Spain

Sep 2015 to Mar 2020

- Dissertation: Water Societal Metabolism in Yucatan Peninsula, Mexico. Application of Multi-Scale Integrated Analysis of Societal and Ecosystem Metabolism (MuSIASEM)
- Three peer-reviewed publications as first author
- Built exploratory tool to analyse effect of climate change in water demand (R, QGIS, Tableau)

Master's degree in Sustainability Science and Technology

Universitat Politècnica de Catalunya, Institute for Sustainability Science

Barcelona, Spain

- 120 ECTS. Grade: 8.9/10 Rank 2/12
- MSc thesis: Societal Metabolism in the Energy Sector in Catalunya, applying spatial analysis with QGIS
- Collaborated with multi-cultural and interdisciplinary team

Master's degree in Logistics

Universidad de Zaragoza, Zaragoza Logistics Center

Zaragoza, Spain

- 105 ECTS. Grade: 9.1/10
- MSc thesis: Methodologies for a Sustainability Balanced Scorecard in the Logistics Sector
- Professional fellowship program in logistics. Training program in multiple companies

Undergraduate degree in Industrial Engineering

Instituto Tecnológico de Mérida

Mérida, México

- Grade: 9.4/10

Selected publications

Peer reviewed

- **Rodríguez-Huerta, E.**, Bell, B., Jackson, B., Blackstone, N.T., Battaglia, K., Marquez, A.S., Benoît Norris, C., Decker Sparks, J.L., Conrad, Z., Matteson, J. (anticipated 2025). The human cost of current and recommended diets in the U.S. (*accepted for Nature Food*). <https://doi.org/10.21203/rs.3.rs-4999594/v1>
- **Rodríguez-Huerta, E.**, Leão L., Landman, T. (2025). Climate change, decent work and workers' health in Brazil: theoretical considerations. *Revista Brasileira de Saúde Ocupacional* <https://doi.org/10.1590/2317-6369/16224en2025v50eddsst12>

- Tigchelaar, M., Jackson, B., Selig, E., Davis, A., O'Regan, E., Trond, K., Nakayama, S., Boyd, D., Cheung, W., **Rodríguez-Huerta**, E., Williams, C., Decker Sparks, J. (2025). Conceptualization of Decent Work in Fishing in a Changing Climate. *Marine Policy*. <https://doi.org/10.1016/j.marpol.2025.106846>
- Boyd, D., Jackson, B., Decker Sparks, J., Giles M. F, Girindran, R., Gosiling, S., Trodd, Z. Ni Bhriain, L., **Rodríguez-Huerta**, E. (2024). The future of decent work: Forecasting heat stress and the intersection of sustainable development challenges in India's brick kilns. *Sustainable Development*, <https://doi.org/10.1002/sd.3272>
- Blackstone, N.T., Battaglia K., **Rodríguez-Huerta**, E., Bell, B., Decker Sparks J., Cash S., Conrad Z., Nikkhah A., Jackson B., Matteson J., Gao S., Fuller K., Zhang F.F., Webb P. (2024). Diets cannot be sustainable without ensuring the well-being of communities, workers, and animals in food value chains. *Nature Food*, <https://doi.org/10.1038/s43016-024-01048-0>
- Lumley-Sapanski A., **Rodríguez-Huerta**, E., Young, M., Nicholson A., Schwarz K. (2024). Criminalizing survivors of modern slavery: the United Kingdom's National Referral Mechanism as a border-making process. *Journal of Social Policy*, <https://doi.org/10.1017/S0047279424000230>
- Blackstone, N.T., **Rodríguez-Huerta**, E., Battaglia K., Jackson B., Jackson E., Benoît Norris C., Decker Sparks, J.L. (2023). Forced labour risk is pervasive in the US land-based food supply. *Nature Food*. <https://doi.org/10.1038/s43016-023-00794-x>

Non-Peer reviewed

- Cockayne, J., **Rodríguez-Huerta** E., Burcu, O. (2022). The Energy of Freedom? Solar Energy, Modern Slavery and the Just Transition. *Research report*, 70,000 words, funded by the British Academy: <https://www.thebritishacademy.ac.uk/publications/the-energy-of-freedom-solar-energy-modern-slavery-and-the-just-transition/>
- Boyd, D., **Rodríguez-Huerta**, E., Jackson B., Decker Sparks, J.L. (2021). The Social and Ecological Impacts of Supply Chains. *Research report*, 16,000 words, funded by the World Wide Fund for Nature.

Under review/Development

- **Rodríguez-Huerta**, E., Jackson, B., Blackstone, N.T., Decker Sparks, J.L., (anticipated 2025). Modern slavery's carbon cost in supply chains: the case of Brazilian soy and sugarcane. *Book chapter for Supply Chains and the SDGs (Elgar Companion)*.
- **Rodríguez-Huerta**, E., Trevizan, A., Landman, T., et al. (anticipated 2026). Exploratory Analysis of the interrelationships between occupational health and decent work (*in preparation for World Development*).
- **Rodríguez-Huerta**, E., Walker, R., Ni Bharain, L., Boyd, D., Jackson, B., (anticipated 2026). Exploration of off-season labour patterns for Indian brick kiln workers (*in preparation for International Journal of Operations & Production Management*).
- Bell, B., Cudhea, F., Battaglia, K., Marquez, A.S., Morris, E., Nelson, J., **Rodríguez-Huerta**, E., Jackson, B., Wang, L., Conrad, Z., Decker Sparks, J.L., Luo, H., Fuller, K., Gao, S., Harney, B., Cash, S., Webb,P., Zhang, F.F.,Blackstone, N.T. (anticipated 2026). Shifting to recommended diets in the US improves health outcomes, but has differing environmental, social, and economic impacts. (*in preparation for Nature Food*).
- Jackson, B., **Rodríguez-Huerta**, E., Marshall, H., Pereira, A.C., Chandler, C., Light, M., Iqbal, S., Boyd, D., Decker Sparks, J.L. (anticipated 2026). Measuring the co-occurrence of tree loss and modern slavery in Brazil between 2001-2021 (*in preparation TBD*).

Other selected outcomes

- **Rodríguez Huerta**, E., (2024-ongoing). Multilingual website for dissemination activities and results of the project 'Climate Change, Occupational Health, and Decent Work: Worker Vulnerabilities and Responses in Brazilian Agriculture' *Digital Artefact: Website Content* <https://www.clidewo.com>
- Jackson, B., **Rodríguez Huerta**, E., Boyd, D., Girindran, R., Ni Bhriain, L., & Trodd, Z. (2024). Environmental Impacts Training Module: Assessing Risks to Workers in India (Versions: English, Hindi, Punjabi & Bengali) *Digital Artefact: Website Content* <https://nottingham-repository.worktribe.com/output/44691673>

- **Rodriguez Huerta, E.,** (2023). Interactive Visualizations to expand on results from the ‘Forced labor risk is pervasive in the US land-based food supply (Blackstone et al. 2023). *Digital Artefact: Website Content* <https://sites.tufts.edu/lasting/data/>

Grants and Fellowships

UK Home Office (Modern Slavery Innovation Fund) The GeoAI for Decent Work: Innovating Engagement with Government, Civil Society and Survivors	£322,000 (Co-I) 2025 - 2026
British Academy (ISPF ODA Challenge-Oriented Research Grants) Climate Change, Occupational Health, and Decent Work: Worker Vulnerabilities and Response in Brazilian Agriculture	£178,000 (PI) 2024 - 2025
UK Home Office (Modern Slavery Innovation Fund) Adding value to the GeoAI Platform for Decent Work: A Survivor-Informed Socio-Ecological Assessment	£149,000 (Co-I) 2024 - 2025
Industry collaboration (Retail sector) Salient and Evolving Human Rights Risks	£450,000 (Co-I) 2024 - 2025
UK Home Office (Modern Slavery Innovation Fund) Using Data Science to Transform India's Brick Kilns	£799,000 (Co-I) 2024 - 2025
British Academy (Just Transitions within Sectors and Industries Globally) Solar Energy, Modern Slavery, and the Just Transition	£80,000 (Co-I) 2021 - 2022
World Wild Fund (USA) The Social and Ecological Impacts of Supply Chains	£60,000 (Co-I) 2021 - 2022
Consejo Nacional de Ciencia y Tecnología (CONACyT - México) PhD and MSc Scholarship	£80,000 (Scholarship) 2014 - 2020
Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR) MSc Fellowship for Directed Academic Activities	£4,000 (Scholarship) 2014 - 2015

Awards

Instituto Tecnológico de Mérida Distinguished alumni of 2025 in the Industrial Engineering programme	2025
THE Awards (Finalist, Research Project of the Year in Social Sciences) Team member of Slavery from Space: Using Satellites for Human Rights and Sustainable Development	2024
University of Nottingham Knowledge Exchange and Impact Awards 2024 Team member of Slavery from Space: Using Satellites for Human Rights and Sustainable Development	2024
Chamber of Commerce of Aragon, Spain (Memoria Pilot award) Report in Logistics Excellence for large enterprises (La Bella Easo, Zaragoza)	2008

Selected Events and Collaborations

Conference: II International Congress on Ecological Humanities The Johns Hopkins University - Universitat Pompeu Fabra Public Policy Center • Decent Work and Workers Health nexus in Brazilian agriculture	Barcelona, Spain Jul 2025
Workshop: Just transitions in food, farming and fisheries Ethical Trading Initiative, IIED, BananaLink	London, UK Jun 2025
Workshop: Addressing Human Trafficking and Forced Labour in the Anthropocene Survivor Alliance, The Human Trafficking Legal Center, The Freedom Fund	Oxford, UK Apr 2025
Workshop: Visualizing Complexity Science Complexity Science Hub	Vienna, Austria Sep 2024
Collaboration: Friedman School of Nutrition Science and Policy at Tufts University Leading A Sustainability Transition in Nutrition Globally (LASTING) Project	Boston, MA, USA 2022 to 2024
Workshop: Decent Fisheries Work in a Changing Climate Stanford University, the King Center and Center for Ocean Solutions	Stanford, CA, USA May 2023
Brown bag sessions	Stanford, CA, USA

- Talk: How Social Life Cycle Assessment (S-LCA) can help address forced labour

22nd Ordinary Session of the Yucatan Peninsula Basin Council

Mérida, México

Consejo de Cuenca de la Península de Yucatán & Ithaca Environmental

Feb 2021

- Invitation to co-design the Regional Water Program 2020-2024 of the Yucatan Peninsula

Teaching

Graduate

Data management and visualization for academic research

Mérida, México

Instituto Tecnológico de Mérida

Nov 2021

- Course for MSc students in Regional Development Studies

Seminars

Seminar: Sustainability science to tackle forced labour: methods and measurements

Barcelona, Spain

Universitat Politècnica de Catalunya, Institute for Sustainability Science

Jul 2025

Selected extra training**Research Communication Programme**

Research Academy University of Nottingham

Jun 2025

MOOC: Partnering for change: Link research to societal challenges

Network for Transdisciplinary Research td-net of the Swiss Academies of Arts and Sciences

Apr 2025

Stepping into Leadership 2023/24. LMA Development Programmes

Research Academy University of Nottingham

Feb 2024

MOOC: Analysing/Presenting Data/Information

Edward Tufte Course (8h)

Oct 2023

MOOC: Data Science: Statistics and Machine Learning Specialization

Coursera Programme (Johns Hopkins University) (133h)

Jan 2021

MOOC: Specialization in Data Science (R Programming)

Coursera Programme (Johns Hopkins University) (290h)

Oct 2020

MOOC: Spatial Data Science and Applications

Coursera Programme (Yonsei University) (24h).

Sep 2020

MOOC: Systems Thinking in Public Health

Coursera Programme (Johns Hopkins University) (20h)

Aug 2020

MOOC: Ecosystem Services.- A Method for Sustainable Development

Coursera Programme (University of Geneva) (24h)

Jul 2020

MOOC: Sustainability of Social-Ecological Systems: Nexus among Water, Energy, and Food

Coursera Programme (Universitat Autònoma de Barcelona) (48h)

Jul 2020

Skills and Languages

Languages

Spanish (Native), English (Fluent)

Programming and Markdown language

R, SQL (intermediate), Quarto

Software

*Tableau [Desktop and Prep], Rstudio, QGIS, OpenLCA, Zotero, VosViewer, Xerte (intermediate), Affinity (learning),**Observable (learning)***Academic Service**

Grants and Journals Reviewed

British Academy for Knowledge frontiers: Just Transition, Journal of Cleaner Production, Energy Policy, Hydrogeology Journal, Joule, Anti-Trafficking Review, Economía Creativa.

Student Supervision

- PhD Students: 3 supervised – Topics in Food Systems
- MSc Students: 4 supervised – Research in data extraction, and regional development
- Support for MSc students from the Technical University of Denmark (DTU) for the project “Sociotechnical dimension of renewable energies” (Apr 2024).
- BSc Students: 3 supervised – Placement students related to data science projects and just transition

Societies

- Member of The British Academy Career Researcher Network since 2023.